

时间同步接口

来源: https://support.unitree.com/home/zh/A2_SDK_Development_Guide/time_sync_interface

时间同步接口

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A2的PC1内部内置了时间同步源，每次将A2设置为WiFi模式时，该时钟源都会进行网络授时。您可以使用Chrony（推荐）或Systemd-Timesyncd来与A2的PC1进行时钟同步。

时钟同步带来的波动

在PC1进行网络授时同步时，可能会导致短时间的时间波动，故推荐您在使用依赖系统时间的程序时关闭A2的WiFi模式。

安装与配置Chrony时钟源

首先需要安装Chrony

```
sudo apt-get install chrony
```

修改配置文件/etc/chrony/chrony.conf，增加如下配置：

```
# 使用内网 NTP 服务器
server 192.168.123.161 iburst prefer

# 如果启动时网络未就绪，允许后续校时
makestep 1.0 3

# 允许硬件时钟同步
rtcsync

# 日志（调试时很有用）
log tracking measurements statistics

logdir /var/log/chrony
```

```
#
# About using servers from the NTP Pool Project in general see (LP: #104525).
# Approved by Ubuntu Technical Board on 2011-02-08.
# See http://www.pool.ntp.org/join.html for more information.
pool ntp.ubuntu.com iburst maxsources 4
pool 0.ubuntu.pool.ntp.org iburst maxsources 1
pool 1.ubuntu.pool.ntp.org iburst maxsources 1
pool 2.ubuntu.pool.ntp.org iburst maxsources 2

# This directive specify the location of the file containing ID/key pairs for
# NTP authentication.
keyfile /etc/chrony/chrony.keys

# server

server 192.168.123.161 iburst prefer

# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift

# Uncomment the following line to turn logging on.
log tracking measurements statistics

# Log files location.
logdir /var/log/chrony

# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0

# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
rtcsync

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3
```

重启Chrony，验证同步状态

```
sudo systemctl restart chrony
```

```
chronyc sources -v
```

在重启Chrony这一步，概率会遇到此问题：

```
(base) supportcbh@supportcbh-MDF-XX:~$ sudo systemctl status chrony.service
● chrony.service – chrony, an NTP client/server
   Loaded: loaded (/lib/systemd/system/chrony.service; enabled; vendor preset:
   enabled)
   Active: failed (Result: core-dump) since Wed 2026-01-07 20:24:23 CST; 5s ago
     Docs: man:chronyd(8)
           man:chronyc(1)
           man:chrony.conf(5)
   Process: 36514 ExecStart=/usr/lib/systemd/scripts/chronyd-starter.sh $DAEMON_OPTS
   (code=exited, status=0/SUCCESS)
   Process: 36522 ExecStartPost=/usr/lib/chrony/chrony-helper update-daemon
   (code=exited, status=0/SUCCESS)
   Main PID: 36519 (code=dumped, signal=SYS)

1月 07 20:24:23 supportcbh-MDF-XX systemd[1]: Starting chrony, an NTP client/server...
1月 07 20:24:23 supportcbh-MDF-XX chronyd-starter.sh[36516]: WARNING: libcap needs an
update (cap=40 should have a name).
1月 07 20:24:23 supportcbh-MDF-XX chronyd[36519]: chronyd version 3.5 starting (+CMDMON
+NTP +REFCLOCK +RTC +PRIVDROP +SCFILTER +SIGND +ASYNCDNS +SECHASH +IPV6 -DEBUG)
1月 07 20:24:23 supportcbh-MDF-XX chronyd[36519]: Loaded seccomp filter
1月 07 20:24:23 supportcbh-MDF-XX systemd[1]: Started chrony, an NTP client/server.
1月 07 20:24:23 supportcbh-MDF-XX systemd[1]: chrony.service: Main process exited,
code=dumped, status=31/SYS
1月 07 20:24:23 supportcbh-MDF-XX systemd[1]: chrony.service: Failed with result 'core-
dump'.
```

Chrony调用了被系统安全过滤器禁止使用的指令。解决方案：禁用Seccomp 过滤。修改/etc/default/chrony可修正此问题。

```
DAEMON_OPTS="-F 0"
```

要验证Chrony是否与PC1的同步源已建立同步，使用此命令验证。

```
chronyc sources -V
```

```
chronyc tracking
```

```
(base) supportcbh@supportcbh-MDF-XX:~$ chronyc sources -V
210 Number of sources = 9
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 192.168.123.161            3    6   377    2  +4881ns[+6893ns] +/-   26ms
^- alphyn.canonical.com      2    7   175   62  +105ms[ +105ms] +/-  287ms
^- prod-ntp-5.ntp4.ps5.cano> 2    6   377   63   +31ms[  +31ms] +/-  153ms
^- prod-ntp-3.ntp1.ps5.cano> 2    7   377   67   +66ms[  +66ms] +/-  185ms
^- prod-ntp-4.ntp1.ps5.cano> 2    7   377   68  +2216us[+2218us] +/-  125ms
^? stratum2-1.ntp.mow01.ru.> 2    7    1  205  +6641us[+6603us] +/-  169ms
^- 111.230.189.174           2    7   177  134  -2436us[-2450us] +/-   38ms
^- ip212-227-240-160.pbiaas> 2    7   357    1   -14ms[  -14ms] +/-  187ms
^C^- 23.157.160.168           2    7   155    6   +22ms[  +22ms] +/-  192ms
(base) supportcbh@supportcbh-MDF-XX:~$ chronyc tracking
Reference ID      : C0A87BA1 (192.168.123.161)
Stratum          : 4
Ref time (UTC)   : Wed Jan 07 12:53:46 2026
System time      : 0.000007202 seconds slow of NTP time
Last offset      : +0.000002012 seconds
RMS offset       : 0.000158323 seconds
Frequency        : 15.779 ppm fast
Residual freq    : -0.000 ppm
Skew             : 0.298 ppm
Root delay       : 0.034696419 seconds
Root dispersion  : 0.007184769 seconds
Update interval  : 65.2 seconds
Leap status      : Normal
```

安装配置Systemd-Timesyncd时钟源

Chrony和Systemd-Timesyncd的互斥关系

chrony和systemd-timesyncd互斥，安装chrony会自动卸载systemd-timesyncd。使用apt install可以重新换回Systemd-Timesync时钟源（不推荐）

```
sudo systemctl unmask systemd-timesyncd
sudo apt install systemd-timesyncd
```

```
sudo systemctl enable --now systemd-timesyncd
```

编辑配置文件/etc/systemd/timesyncd.conf为以下内容：

```
[Time]
NTP=192.168.123.161
FallbackNTP=
PollIntervalMinSec=32

PollIntervalMaxSec=2048
```

```
GNU nano 4.8 /etc/systemd/timesyncd.conf
# This file is part of systemd.
#
# systemd is free software; you can redistribute it and/or modify it
# under the terms of the GNU Lesser General Public License as published by
# the Free Software Foundation; either version 2.1 of the License, or
# (at your option) any later version.
#
# Entries in this file show the compile time defaults.
# You can change settings by editing this file.
# Defaults can be restored by simply deleting this file.
#
# See timesyncd.conf(5) for details.
[Time]
NTP=192.168.123.161
FallbackNTP=
#RootDistanceMaxSec=5
PollIntervalMinSec=32
PollIntervalMaxSec=2048
```

随后，启用并重启服务

```
sudo systemctl enable systemd-timesyncd
```

```
sudo systemctl restart systemd-timesyncd
```

验证同步情况

```
(base) supportcbh@supportcbh-MDF-XX:~$ timedatectl timesync-status
    Server: 192.168.123.161 (192.168.123.161)
Poll interval: 1min 4s (min: 32s; max 34min 8s)
    Leap: normal
    Version: 4
    Stratum: 3
    Reference: CB6B0658
    Precision: 1us (-22)
Root distance: 22.734ms (max: 5s)
    Offset: +96us
    Delay: 690us
    Jitter: 0
    Packet count: 1
    Frequency: -15.286ppm
(base) supportcbh@supportcbh-MDF-XX:~$ timedatectl status
    Local time: == 2026-01-07 21:05:11 CST
    Universal time: == 2026-01-07 13:05:11 UTC
    RTC time: == 2026-01-07 21:05:11
    Time zone: Asia/Shanghai (CST, +0800)
System clock synchronized: yes
    NTP service: active
    RTC in local TZ: yes

Warning: The system is configured to read the RTC time in the local time zone.
This mode cannot be fully supported. It will create various problems
with time zone changes and daylight saving time adjustments. The RTC
time is never updated, it relies on external facilities to maintain it.
If at all possible, use RTC in UTC by calling
'timedatectl set-local-rtc 0'.
```